



Thesis Proposals in Artificial Intelligence for Cutaneous Lymphoma Research

Overview

Cutaneous lymphomas are rare cancers of the skin. The most common type, mycosis fungoides, is difficult to recognise in its early stages because it looks very similar to common inflammatory skin diseases such as eczema and psoriasis, and the literature reports that a confident diagnosis often requires repeated biopsies over a long period. This makes the disease an interesting and demanding field for artificial intelligence, where the value of a model depends not on a single accuracy figure but on whether it holds up on data from a different source, on different skin tones and on the difficult cases that matter in practice.

The topics below are offered in collaboration with researchers from the field of medicine, who contribute the clinical and biological expertise, help to define the diagnostic standard and take part in interpreting and writing up the results. Each topic combines a **theoretical component** (analysis, study, comparison) with a **practical component** (data work, experiments and implementation).

Three points apply to all the topics. First, every topic is designed to be completed in full with publicly available data and published literature, so that no dissertation depends on access to a hospital. Second, students never work with identifiable patient data. Where a clinical partner is available, a topic can be extended with an evaluation on clinical data: this follows the approval procedures of the clinical site and of the College, the analysis runs at the clinical site, and the student receives only anonymised values and aggregate results. Third, the topics are parts of one research programme: they follow common working rules and a shared format for data and results, so that the findings of the individual dissertations can be brought together into a larger study, as described at the end of this document.

Links and Resources

Open Science Framework	https://osf.io/
PROSPERO register of reviews	https://www.crd.york.ac.uk/prospERO/
PRISMA reporting guidelines	https://www.prisma-statement.org/
EQUATOR Network reporting guidelines	https://www.equator-network.org/
TRIPOD reporting guideline	https://www.tripod-statement.org/
PROBAST risk of bias tool	https://www.probast.org/
Rayyan screening tool	https://www.rayyan.ai/
Fitzpatrick17k image list	https://github.com/mattgroh/fitzpatrick17k
Image and annotation datasets	Fitzpatrick17k (Groh et al., 2021), Diverse Dermatology Images (Daneshjou et al., 2022) and SkinCon (Daneshjou et al., 2022), obtained under their own terms of use



Thesis Topics

The topics are presented in order of technical difficulty, starting with the most demanding in terms of programming and machine learning.

1. Deep Learning for Recognising Mycosis Fungoides Among Its Inflammatory Mimics

This topic develops image classification models that separate mycosis fungoides from the inflammatory skin diseases that resemble it, and then tests honestly how far those models generalise. It is the most demanding of the topics and suits a student who is confident with Python and deep learning, and who has access to a GPU or is willing to work with cloud resources. The scientific interest lies less in reaching a high score on a familiar test set and more in measuring what happens when the images come from a different atlas, when the skin tone changes, and when the cases are the genuinely difficult ones.

Theoretical component

The student studies transfer learning and the recent generation of foundation models for dermatology, and the reasons why medical image models often perform worse outside the dataset on which they were developed. The study covers dataset shift and shortcut learning, where a model learns an artefact such as a watermark or a ruler instead of the disease, fairness and performance differences across skin tones, the calibration of predicted probabilities, and methods for explaining predictions. It also covers the established reporting standards for prediction models, which define what a credible development and validation study must report.

Practical component

The student assembles a development set of clinical photographs from public dermatology datasets. The Fitzpatrick17k collection alone lists about 180 photographs labelled as mycosis fungoides, taken from two online atlases, together with several hundred photographs of psoriasis, eczema, lichen planus and other diseases that resemble it. The medical collaborators choose which inflammatory diseases form the comparison group and, because the public labels say nothing about the stage of the disease, mark which photographs show early disease. Careful data preparation follows: duplicate and near duplicate images are found with perceptual hashing and embedding similarity and grouped together, the splits are made at group level so that near identical images never appear in both training and testing, and watermarks and borders are removed so that the model cannot use the source of the image as a clue.

The student then trains and compares several approaches, typically a convolutional network pretrained on natural images, a self supervised vision transformer used both as a frozen feature extractor and with partial fine tuning, and at least one dermatology foundation model. Training uses cross validation, class weighting, several random seeds, and a decision threshold that is fixed in advance from the cross validation predictions with a clinically sensible objective, such as a minimum sensitivity. Probabilities are calibrated, and the full analysis plan is written down before any external test set is opened.

Evaluation is the heart of the topic. The models are trained on images from one atlas and tested on images from another, results are reported with confidence intervals and broken down by skin tone and by type of mimicking disease, and calibration is reported alongside discrimination. The student then examines what the models actually use, with saliency maps and with the clinical concept annotations available for these datasets, and analyses the errors together with the medical collaborators.



If a clinical partner is available and the necessary approvals are in place, the work can be extended with a real world external validation. The student prepares a self contained software package with fixed versions and final model weights, tests it first on public images, and provides it to the clinical site, where it runs on anonymised photographs from routine care. Only predictions with anonymous codes and aggregate metrics are returned to the student.

Main outcomes and further work

The main outcomes are a reproducible comparison of modern image models on this task, an external evaluation across image sources with subgroup and calibration results, and an explainability and error analysis. Together these form a complete study without any clinical data. The dissertation can be extended with the clinical evaluation described above, and with a reader study in which clinicians assess the same images with and without model support.

Required skills: strong Python and PyTorch, experience with training and evaluating neural networks, and familiarity with basic statistics. No medical background is required, as the clinical knowledge is provided by the collaborators.

2. Can General Purpose Vision Language Models Recognise Mycosis Fungoides?

Large vision language models can describe a photograph and suggest a diagnosis without any task specific training, and dermatology foundation models can be used in the same way. How well they separate a rare lymphoma from the common diseases that resemble it has received little attention so far, and establishing what has been published is the first step of the work. This topic evaluates such models as they are, without training a large model, so it depends on careful experimental design rather than on computing power. It suits a student who programs comfortably in Python and is interested in how models should be evaluated.

Theoretical component

The student studies how contrastive image and text models and generative vision language models work, how they are used for classification without training, and how strongly their answers depend on the wording of the prompt. The study covers the repeatability of generated answers, the measures of diagnostic accuracy and their confidence intervals, performance differences across skin tones, and the problem of contamination, where a benchmark looks better than it should because the test images were already part of the data on which a model was trained.

Practical component

The student works on the same public photographs of mycosis fungoides and its mimics as topic 1, with the medical collaborators choosing the comparison diseases. The protocol is written and registered before any model is run. It fixes the models to be compared, which include several openly available vision language models and at least one dermatology foundation model, the exact prompts, the number of repeated runs, and the two tasks: a forced choice among a list of diagnoses, and an open differential diagnosis in which the three most likely diseases are requested.

The student reports the sensitivity and specificity for mycosis fungoides, how often it appears among the three suggested diagnoses, the agreement between repeated runs, and the same results broken down by skin tone and by mimicking disease. A simple classifier trained on frozen image embeddings serves as a reference point. The medical collaborators review the

explanations that the models give for a sample of cases, and judge whether the clinical features they mention are really visible in the photograph. Because the photographs come from public atlases, the student discusses the risk of contamination openly and, where possible, repeats the test on images from a source that models are less likely to have seen.

Main outcomes and further work

The main outcomes are a registered evaluation protocol, a comparison of the models with confidence intervals and subgroup results, and an analysis of their errors and explanations with medical experts. The dissertation can be extended with a study of prompting strategies, with a small number of example images given to the model, and with a comparison against the trained models of topic 1.

Required skills: good Python, experience with calling models through libraries or programming interfaces, and basic statistics. Modest computing resources are enough.

3. Cutaneous Lymphoma in Public Dermatology Image Datasets: A Data Audit

Every image model in this field depends on the photographs it learns from. Earlier work has examined how mycosis fungoides appears in online image searches, but the datasets used to train models have not, to our knowledge, been audited for this disease, and confirming this is the first step of the work. This topic documents how well cutaneous lymphomas are represented in public dermatology image datasets, how reliable the labels are and which skin tones are missing. It suits a student who is careful with data and prefers analysis to model building.

Theoretical component

The student studies how datasets are documented and audited, including datasheets for datasets, label noise, duplicate images and the leakage they cause between training and testing, and the scales used to describe skin tone. The study also covers representation bias and its effect on the performance of models, and the statistics of agreement between raters.

Practical component

The student builds an inventory of public dermatology image datasets and atlases with their terms of use and records, for each one, how many images are labelled as mycosis fungoides or another cutaneous lymphoma, where the images come from, which skin tones they show and what other information is available. As an indication, the Fitzpatrick17k list contains about 180 images labelled as mycosis fungoides, of which fewer than twenty show the two darkest skin types. Duplicates within and across datasets are found with perceptual hashing and embedding similarity.

The labels are then checked by the medical collaborators, who review the lymphoma images without seeing the original label and record whether the diagnosis is plausible, the quality of the photograph, the type of lesion (patch, plaque or tumour) and the body site. A second rater reviews a subset, and the agreement between the two is reported. The result is an annotation table that refers to the images by the identifiers of the original datasets, so that no image is redistributed, together with recommended training and test splits that are free of duplicates.

Main outcomes and further work

The main outcomes are a documented audit of how cutaneous lymphomas are represented in public image datasets and an openly released annotation table with a permanent identifier, which other researchers can use and on which topics 1 and 2 build. The same audit can be repeated for other rare skin diseases by later dissertations, and it can be extended with an experiment that shows how much the label errors and duplicates change the measured performance of a model.

Required skills: Python for data handling, patience and attention to detail, and basic statistics. Little machine learning is needed.

4. Language Models as a Source of Patient Information on Cutaneous Lymphoma, in English and in Greek

People with a rare disease increasingly ask conversational language models to explain their diagnosis and treatment. This topic measures how accurate, complete, safe and readable the answers are for cutaneous lymphomas, in English and in Greek, as judged by medical experts against current clinical guidelines. A preliminary search found very few published evaluations of language models for this disease. No patients and no patient data are involved, which makes this the quickest topic to complete.

Theoretical component

The student studies how language models generate text and why they can state something false with confidence, and the instruments used to assess the quality of health information for patients. The study covers the measurement of readability and how far the usual formulas are valid for Greek, and the design choices that decide whether such a study is credible: the wording of the questions, repeated runs, recording of model versions and dates, blinding of the raters and agreement between them.

Practical component

Together with the medical collaborators, the student builds a set of questions that patients ask about diagnosis, staging, treatment, prognosis and daily life, drawn from clinical guidelines and from the information published by patient organisations, and prepares it in both languages. The protocol, including the rating scales, is registered before any answer is collected. Each question is put to several widely used language models, more than once and with fixed settings.

The answers are stripped of anything that identifies the model, placed in random order and rated independently by at least two medical raters for accuracy, completeness and potential for harm. The student computes readability, compares the models and the two languages, reports the agreement between raters, and analyses the kinds of error that occur, such as outdated treatment recommendations or advice that could delay care.

Main outcomes and further work

The main outcomes are a registered protocol, an openly available set of questions and ratings, and a comparison of models and languages. The same protocol can be repeated for other diseases and languages by later dissertations, so that the results accumulate. For a student who wants a stronger technical component, the dissertation can be extended by giving the models access to the text of the clinical guidelines and measuring whether the answers improve.



Required skills: basic Python scripting, basic statistics, organised work, and fluent Greek and English.

5. How Trustworthy Are Published Artificial Intelligence Models for Cutaneous Lymphoma? A Methodological Quality Appraisal

Published studies report encouraging accuracy for models that diagnose or predict the course of cutaneous lymphomas, but an accuracy figure means little without knowing how the model was validated. This topic appraises the published prediction models in the field with the tools designed for that purpose, one for risk of bias and one for completeness of reporting. A preliminary search of PubMed in October 2026 found about forty studies that apply artificial intelligence to cutaneous lymphomas and no published appraisal of this kind. The topic suits a student who understands how machine learning models are evaluated and enjoys reading research critically.

Theoretical component

The student studies the methodology of prediction model research: the difference between developing and validating a model, internal and external validation, overfitting, leakage between training and test data, splitting at the level of the patient, and calibration. The study covers the PROBAST tool and its domains, the TRIPOD reporting guideline, their recent versions for artificial intelligence, and the way in which two reviewers carry out and report such an assessment.

Practical component

The student writes and registers a protocol, then runs a documented search across the main biomedical and computer science databases. Where topic 6 runs in the same year, the two dissertations share the search and remain separate in everything else. Each included study is assessed independently by two reviewers, the student and the supervisor or a medical collaborator, after a pilot round on a small sample to align judgements.

For every study the student records the risk of bias in each domain and the reporting items that are present or missing, and extracts how the data were split, whether an external validation was performed, the sample size and whether code and data are available. The results are presented as summary tables and figures, with an analysis of the weaknesses that occur most often and of how practice has changed over time, and they end in concrete recommendations for future studies in rare skin diseases.

Main outcomes and further work

The main outcomes are a registered protocol, a complete and openly available appraisal of the published models, and a set of recommendations for future studies. The dissertation can be extended by attempting to reproduce the results of one study whose code and data are available.

Required skills: a sound understanding of how machine learning models are trained and evaluated, careful reading, and good academic English. Little programming is needed.

6. Artificial Intelligence in Cutaneous Lymphomas: A Scoping Review with Clinical Expert Consultation

A preliminary search of PubMed in October 2026 found about forty studies that apply artificial intelligence to cutaneous lymphomas, but no systematic or scoping review that maps what it has actually been used for, on which kinds of data, and how close any of it is to clinical use. This topic produces that map with a formal review methodology, and it is the least technical of the topics, suiting a student who is organised, thorough and interested in understanding a research field in depth rather than in writing large amounts of code.

Theoretical component

The student studies the methodology of scoping reviews and how it differs from a narrative literature review and from a systematic review, including the established framework for defining the population, the concept and the context of a review, the corresponding reporting checklist, and the practice of registering a protocol before the work begins. The study also covers the criteria by which prediction models are judged, such as how they were validated, whether the data were split at patient level, whether probabilities were calibrated and whether the model was compared with clinicians, which become the analytical lens of the review.

Practical component

The student writes a protocol and registers it on the Open Science Framework, since the PROSPERO register does not accept scoping reviews, after checking that no identical review is already under way. The student then builds and documents a search strategy across the main biomedical and computer science databases, with backward and forward citation searching. Screening is done by two independent reviewers, the student and a medical collaborator, first on titles and abstracts and then on full texts, beginning with a pilot round on a small sample to align judgements, with disagreements resolved by the supervisor.

For every included study the student extracts a structured set of fields agreed in advance: the disease and the clinical task, the type of data, the sample size and its source, the reference standard, the model and the type of validation, the reported metrics, whether calibration and comparison with clinicians were reported, and whether code and data are available. The results are synthesised into a map of data types against clinical tasks and into a table that places each line of work on a scale of maturity, from proof of concept through internal and external validation to prospective clinical evaluation, with a narrative synthesis of the gaps.

The review closes with a consultation stage, which is the optional final step of the standard scoping review framework. The preliminary findings are presented to a small group of clinicians and researchers, who answer a short anonymous questionnaire on which applications they would actually use, what is missing and what obstacles they foresee. Their responses are reported as a separate part of the results, giving the review a practical perspective that a literature analysis alone cannot provide.

Main outcomes and further work

The main outcomes are a registered protocol, a documented and repeatable search, a complete map of the field with a maturity assessment, and a structured account of clinical priorities. The dissertation can be extended with a deeper comparison between the applications that researchers develop and the needs that clinicians describe.



Required skills: systematic and careful work, good academic English, and basic data handling for tables and figures. Programming experience is useful but not necessary.

How the Topics Fit Together

The six topics are independent dissertations, and each can be carried out and assessed on its own. They are also designed as parts of one research programme on artificial intelligence for cutaneous lymphomas, in which the findings of the individual dissertations add up to a larger study than any single one could deliver. The topics fall into three strands.

The first strand is about images. The data audit (topic 3) provides a cleaned and annotated image set, on which the general purpose models (topic 2) and the trained models (topic 1) are evaluated with the same splits and the same measures, so that their results can be compared directly. The second strand is about the published literature. The scoping review (topic 6) maps what has been done and the quality appraisal (topic 5) judges how well it was done, on the same body of studies. The third strand is about patients. The patient information topic (4) stands on its own, and its protocol can be repeated for other diseases and languages.

To make this possible, all the dissertations follow the same working rules. The plan of each study is written down, and where appropriate registered, before any results are seen. Data and results are kept in a shared format and a common repository. The work follows the reporting guideline for its type of study, such as PRISMA for reviews and TRIPOD for prediction models, and the code and the analysis are kept so that every result can be reproduced. Students are expected to meet regularly with the medical collaborators, who define the diagnostic standard and help to interpret the findings. Each dissertation is written individually and covers the student's own contribution, and where the results of several dissertations are later brought together, the contribution of each student is acknowledged.